GROUNDCONTROL
Nodal Series

Targeted Low-Frequency Control for Small Rooms

You dial in the mix. The low end feels tight. The kick hits right. The bass sits perfectly. Then you play it somewhere else, and it falls apart. The low end disappears. Or it overwhelms everything. The balance is gone. So, you come back to your room… and it sounds fine again.

> *You didn't get the mix wrong. Your room did.*

WHAT YOU'RE HEARING IN THE LOW END ISN'T JUST YOUR MIX, IT'S YOUR ROOM SHAPING IT.

At bass frequencies, your room dominates what you hear. Certain notes build up and ring. Others cancel out. That "tight" low end you're hearing is being boosted, smeared, or masked by the space you're working in. And because it happens every time you listen, your ears adapt. Your decisions follow. Your mixes follow.

> *If the room isn't controlled, neither is your low end.*

THE PROBLEM LIVES WHERE MOST TREATMENT DOESN'T

Below ~120 Hz, your room stops behaving like a space and starts behaving like a resonator.

Bass energy builds between boundaries, reinforcing specific frequencies and sustaining them long after the signal stops. These resonances caused by room modes can ring for over a second, masking detail and distorting balance.

Meanwhile, standard acoustic treatment does very little here.

Foam, panels, and typical "bass traps" work well in the mids and highs. But at low frequencies, they're simply too thin to absorb meaningfully. The result:

- **A room that feels controlled… but only above ~150 Hz**
- **And a low end that remains unpredictable, inconsistent, and misleading**

THIS IS WHY MIXES DON'T TRANSLATE

In most small studios, the critical problems live between **40 and 100 Hz**—the exact range where:

- Kick and bass define the mix
- Translation issues originate
- Conventional treatment is least effective

Fixing this requires treating the frequencies most rooms can't control. That's what the Nodal Series is designed to do.

> *When the low end is finally controlled, it often sounds less impressive at first, that's how you know it's working.*

WHY YOUR ROOM SOUNDS THE WAY IT DOES

Bass frequencies behave differently from everything else in a small room. At low frequencies, roughly below 120 Hz, the room itself becomes a musical instrument. Sounds at certain frequencies bounce back and forth between parallel walls and reinforces itself, building up into sustained resonances called room modes. Some notes ring for a second or more after the music stops. Others are artificially quiet. The result is a frequency response that can vary by 20 dB or more just by moving your head a foot in any direction.

Standard acoustic treatment such as foam panels, hanging baffles, corner traps made from thick mineral wool genuinely works, and you should use it. Above 150–200 Hz porous materials absorb effectively and make a real difference to flutter echo, mid-range coloration, and stereo imaging. If you haven't treated your room at all, start there.

But porous absorption has a hard physical limit in the bass. To absorb sound, a porous material needs to be a meaningful fraction of the sound wavelength in thickness. At 55 Hz the wavelength is over 20 feet. A 4-inch rockwool panel is about 1% of that wavelength — its absorption coefficient at that frequency is essentially zero. To treat 55 Hz with porous material alone you would need panels over 12 inches deep filling most of your corners. Most people don't have the space, and even if they did the mid-range absorption would be excessive.

The result is that most home studios are treated adequately in the mid-range and completely untreated below 80 Hz, the frequency range where the most damaging modal resonances live.

> *In a typical untreated home studio, the room is still ringing audibly at 55 Hz for over a second after the music stops. Your monitors go silent. The room keeps playing.*

42–109 Hz	Up to −51%	7 inches
CONTINUOUS COVERAGE	**DECAY REDUCTION**	**UNIT DEPTH**
the full home studio problem range	*at 55 Hz in reference room*	*three units fill an 8 ft ceiling*

A DIFFERENT APPROACH

The Nodal Series is designed specifically for this problem. Instead of relying on thickness, it uses **tuned resonance**, targeting the exact frequencies where small rooms fail. The result is controlled decay, reduced modal ringing, and a low end that reflects what's actually in your mix, not what your room is adding to it.

Each unit contains one or two thin panel membranes mounted in front of a sealed air cavity. When bass at the target frequency strikes the membrane, the membrane flexes and vibrates, converting that acoustic energy into heat through internal damping material. The unit absorbs specifically at the frequency it's tuned to, efficiently and in a 179 mm total depth that fits against your wall without eating your room.

The Nodal Series brings that engineering to the home studio in a product designed around the frequencies that cause the most problems in typical small rooms and sized to fit in the corners where those problems are worst. It’s designed to work alongside your broadband treatment, not instead of it. Porous panels handle everything above 120 Hz. The Nodal Series handles the range below 110 Hz that porous panels cannot reach. The two approaches are complementary, each doing the job the other cannot.

> *The question is not whether to use broadband treatment, you should. The question is whether your specific room has problems below 80 Hz that only a tuned absorber can fix. If it does, few solutions at a practical depth can solve it.*

What That Means in Practice

When the low end is controlled:

- You stop overcorrecting with EQ
- Kick and bass sit naturally in the mix
- Decisions translate outside your room
- What sounds right… actually is right

Why corners?

Bass modes build up pressure at room boundaries which are the walls, floor, and ceiling. At a corner where two walls and the floor all meet, that pressure is at its absolute maximum. A panel absorber placed in a corner is working at the highest possible concentration of bass energy, making it far more effective than the same unit placed mid-wall. Four units in corners do more than eight units spread around the walls.

Placement matters more for tuned absorbers than for porous panels. A tuned unit works best at the pressure maximum for its target frequency. The corner is the right location for the same reason it is for any bass treatment: it has the greatest energy concentration regardless of room dimensions.

IS THE NODAL SERIES RIGHT FOR YOUR ROOM?

The Nodal Series makes the most compelling case in rooms where the dominant problems are below 80 Hz. Whether that describes your room depends on your specific dimensions. Rooms in the 14 to 20 feet range typically have their worst axial modes clustering between 40 and 100 Hz, right in the Nodal Series range. Larger rooms push the dominant modes lower; smaller rooms push them higher.

If you have taken RT60 measurements with REW and see elevated decay times between 40 and 100 Hz, the Nodal Series is designed for exactly your problem. If your room's worst decay times are above 100 Hz, broadband porous treatment alone may be sufficient, and is the right first investment regardless.

The simulation at nodal.integeraudio.com lets you enter your room dimensions and see the predicted RT60 before and after Nodal treatment. It is the most direct way to answer whether your room has a problem the Nodal Series can fix.

> *If your room has persistent problems below 80 Hz, the Nodal Series can address them and nothing else at practical depth can. If your room's problems are above 100 Hz, broadband treatment is the right answer.*

THE PRODUCTS

Nodal 55/70 — Dual-Chamber Panel Absorber

The Nodal 55/70 is the foundation of the system. Each unit contains two completely independent absorption chambers — one targeting 55 Hz and one targeting 70 Hz — stacked vertically within a single 630 × 762 mm enclosure. Both chambers share the same 179 mm depth, presenting a flush face to the room with a continuous natural birch plywood frame surrounding two satin black membrane panels.

The 55 Hz chamber uses a mass-loaded panel face to hit the lowest tuning frequency in the system. The 70 Hz chamber uses a lighter panel face at a different internal geometry. Together they provide unbroken coverage from 42 to 86 Hz, cutting decay times by 43 to 51% at these frequencies based on simulations of the reference room configuration.

	Chamber A — 55 Hz	Chamber B — 70 Hz
Tuning frequency	54.6 Hz	70.7 Hz
Coverage (−3 dB)	42–67 Hz	55–86 Hz
RT60: untreated room	1.34 s	0.86 s
RT60: treated (simulation)	0.65 s (−51%)	0.49 s (−43%)
Dimensions (W×H×D)	630 × 762 × 179 mm	—

Overall 630 × 762 × 179 mm (W × H × D) · 7 inches deep

Concept rendering – AI generated

Nodal 85 — Single-Chamber Panel Absorber

The Nodal 85 completes the system. At exactly half the width of the Nodal 55/70, two units sit side by side to form a 630 mm wide pair that stacks seamlessly on top of two Nodal 55/70 units — the same depth, the same frame profile, the same flush face. The Nodal 85 is tuned at 86 Hz, covering 69 to 109 Hz where the system hands off to the range where conventional broadband panels become effective.

In the reference room simulation, four Nodal 85 units at corners reduce RT60 at 85 Hz by 49%, from 0.90 seconds to 0.46 seconds, landing squarely in the target range for critical listening.

Specification	Value
Tuning frequency	86 Hz
Coverage (−3 dB)	69–109 Hz
RT60: untreated room	0.90 s
RT60: treated (ref.)	0.46 s (−49%)
Dimensions (W×H×D)	315 × 762 × 179 mm

WHAT THE SYSTEM COVERS

Used together, the Nodal 55/70 and Nodal 85 provide continuous absorption from 42 to 109 Hz — the full range of the home studio bass problem with no gaps between tuning points. The overlapping coverage bands mean every frequency in this range receives meaningful treatment.

Frequency	Product	Coverage	Before	After	Reduction
55 Hz	Nodal 55/70 Ch. A	42–67 Hz	1.20 s	0.50 s	−58%
70 Hz	Nodal 55/70 Ch. B	55–86 Hz	0.85 s	0.40 s	−53%
85 Hz	Nodal 85	69–109 Hz	0.89 s	0.44 s	−51%
System	2+2 per corner × 4	42–109 Hz	—	—	51–58%

Reference room: 6.10 × 4.88 × 2.44 m (20 × 16 × 8 ft) · 2 × Nodal 55/70 stacked + 2 × Nodal 85 side by side per corner · 4 corners

> *The industry target for critical listening is 0.25–0.55 seconds modal decay time. All three treated values land at or near this target — from a starting point of 0.85 to 1.20 seconds in the untreated room.*

THE CORNER STACK

The Nodal Series is designed around a specific installation: two Nodal 55/70 units stacked vertically in each room corner, with two Nodal 85 units sitting side by side on top. The result is a single 630 mm wide column of treatment that runs from the floor to within 40 mm of a standard 8 ft (2440 mm) ceiling.

This arrangement is deliberate. The two Nodal 55/70 units occupy the lower section of the column, coupling into the floor-corner pressure maximum where 55 and 70 Hz modes concentrate. The Nodal 85 pair sits at the top, coupling into the ceiling corner where the higher frequency modes peak. Both pressure maxima are addressed simultaneously by a single flush column that takes up less than 8 inches of wall depth.

All four units in the stack share the same 179 mm depth and the same frame profile, so every face sits perfectly flush. The 762 mm unit height is chosen so that the full three-unit column reaches exactly 2286 mm, fitting comfortably under a standard 8 ft ceiling with clearance to spare.

Starting with fewer units

If budget requires a phased approach, start with two Nodal 55/70 units at the two front corners behind your monitors. Front corners couple most strongly to the listening position and give you the greatest immediate improvement. Add the Nodal 85 units and rear corner treatment as a second phase.

WHAT TO EXPECT

The first session in a properly treated room often feels unsettling. The low end sounds thinner, less impressive, less present. That's not a problem, it's the point. The resonant reinforcement you were mixing with is gone, and what you're hearing now is closer to what's actually in the recording.

Give it a few sessions. Once your ears adjust to the treated room, you'll find:

- Bass guitar and kick drum sit in the mix without needing heavy corrective EQ. You're hearing them clearly for the first time
- Low end decisions made in the room translate to other systems. The bass that sounds right actually is right
- The frequency balance at the listening position is more consistent, less variation as you move around the room
- Subwoofer integration becomes more predictable. You can hear what it's actually doing

What the Nodal Series does not address:

- Flutter echo and mid-range coloration. Broadband porous panels handle this above 120 Hz and should be part of any complete treatment
- Every bass mode in your room. Other modes in the bass range may still be audible and may benefit from additional broadband treatment at useful depths
- Rooms where the dominant problems are above 100 Hz. If your RT60 measurements show the worst decay times above that range, porous treatment is the right first investment
- Phase issues or stereo imaging problems. These are separate phenomena unrelated to modal decay

> *The goal isn't a room that sounds impressive. It's a room that tells the truth.*

KEY PERFORMANCE RANGE

Frequency: 42–109 Hz
Continuous coverage across the primary small-room problem range

Up to 51% decay reduction
(in modeled reference room conditions)

7-inch depth
Fits into real rooms without consuming space

FINISH & CONSTRUCTION

Every element of the Nodal Series is chosen to look as good as it performs. The enclosure is CNC-cut 12 mm Baltic birch plywood, precise, stable, and distinctive. The membrane panels are painted satin black. The contrast between the natural ply frame and the dark membrane face is the product's visual identity: a warm birch border surrounding each tuned surface, reading as intentional and technical rather than decorative.

Frame & lip	12 mm Baltic birch ply · clear hardwax oil finish · natural grain visible
Membrane panels	MDF · satin black water-based paint · two coats over shellac primer
Reveal	5 mm continuous birch plywood lip around all membrane edges · consistent on all four sides
Crossmember	12 mm total thickness set below membrane panels · 4 mm gap between membranes
Depth	179 mm total · 7 inches
Internal seal	Membrane bonded with 3M 4200 flexible polyurethane adhesive · airtight cavity seal

The birch ply finish is applied before assembly, working on flat components for consistent coverage. The rebate zones are masked during finishing so the adhesive bond between membrane and frame is attached to bare wood. The result is a unit that looks as clean on close inspection as it does from across the room.

FULL TECHNICAL SPECIFICATIONS

Nodal 55/70

Specification	Chamber A — 55 Hz	Chamber B — 70 Hz
Tuning frequency	54.6 Hz	70.7 Hz
Coverage (−3 dB)	42–67 Hz	55–86 Hz
Face panel	MDF + mass-loaded vinyl	MDF
Cavity depth	162 mm	162 mm
Internal damping	Mineral wool 32–45 kg/m³	Mineral wool 32–45 kg/m³
Quality factor Q	2.25	2.25
Overall dimensions (W×H×D)	630 × 762 × 179 mm	—

Nodal 85

Specification	Value
Tuning frequency	86 Hz
Coverage (−3 dB)	69–109 Hz
Face panel	MDF
Cavity depth	164 mm
Internal damping	Mineral wool 32–45 kg/m³
Quality factor Q	2.25
Overall dimensions (W×H×D)	315 × 762 × 179 mm
System coverage (−3 dB)	42–109 Hz (combined with Nodal 55/70)
System RT60 reduction (simulation)	51–58% (reference room · 4+4 corner config.)

ABOUT INTEGER AUDIO

Integer Audio is a boutique electronics and acoustics company. The Nodal Series is designed and built to the same standard as the RMP2 ribbon microphone preamplifier, engineered from first principles, measured, and honest about what it does and doesn't do.

We build tools for people who take sound seriously.

www.integeraudio.com

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USER GUIDE

The Nodal Modal Simulator

Predicting Results Before You Buy

The GroundControl Nodal Series is a precision tool. To ensure you are deploying it effectively, we provide a Modal Simulation tool that calculates the specific resonant behavior of your room.

Step 1: Define Your Room Dimensions

Enter your room's Length, Width, and Height. The simulator will automatically calculate your **Axial Modes**, the frequencies where sound waves bounce between parallel surfaces and create "ringing."

- **Pro Tip:** Look at the **Dominant Mode Decay Summary**. If you see decay times (RT60) above **0.75s** in the 40–100 Hz range, your room is "Boomy" and requires tuned treatment.

Step 2: Place Your Virtual Nodal Units

The simulator allows you to virtually "install" Nodal 55/70 and Nodal 85 units.

- **Corner Loading:** Click the corners on the floor plan. The simulator defaults to corner placement because pressure is highest there, making the resonators most efficient.
- **Stacking:** The Nodal 55/70 is a **dual-chamber unit**; placing one unit automatically provides treatment for both **55 Hz and 70 Hz**. Add multiple units which can be stacked.

Step 3: Analyze the Heatmap

Toggle between the **[Untreated]** and **[Treated]** views.

- **The Heatmap:** Shows "Hot Spots" (High SPL) and "Nulls" (Low SPL).
- **The Goal:** You want to see the "hot" areas (dark reds) become more uniform. A "flatter" heatmap means you can move your head or your chair without the bass response changing drastically.

Step 4: Check the RT60 Improvement

The Simulation provides a calculated **% Improvement**.

- **The Target:** Aim for the **Controlled** range (**0.25 – 0.55s**).
- If your room starts at 1.20s at 56 Hz, adding 4 Nodal units can typically drive that down by **35-50%**, bringing you into the professional reference standard.

Quick Setup Reference

Component	Function
Nodal 55/70	Targets the two deepest "problem" modes in one unit.
Nodal 85	Fills the gap between sub-bass and broadband treatment.
Listening Position	Drag the marker to see the exact frequency response at your desk.
Export Report	Save your configuration to help plan your physical studio layout.